Swastik Ghosh

Bangalore, India. · swastikghosh1106@gmail.com · 9131230644 · **LinkedIn:** swastik-ghosh ,
GitHub: sg1106 , **Portfolio:** LLM Demo

Work Experience

HSBC (Financial Crime and Future Analytics) Bangalore , India
(Manager) Feb, 2026 - present

- Built a multi-agentic GenAI system for EWRA financial-crime analytics, directly connecting to a multi-table database and using GenAI models as tools for schema-aware database understanding, user-query intent classification, and Text-to-SQL generation, followed by automated query execution, data retrieval,mathematical analysis, visualization, and client-ready PDF report generation.
- Developed a multi-agentic RAG system to retrieve relevant data from EWRA reports and perform mathematical operations and contextual analysis using GenAI models as tools.
- Designed and developed an automated web-scraping system that aggregates publicly available internet data to identify financial crime and financial activity related to individuals using name-based searches.
- Developed quantitative analytics and prioritization frameworks for multi-market financial-crime cases across Singapore, UK, and Turkey, supporting risk-based allocation
- Supported trigger generation for financial crime detection, identifying suspicious transactions and flagging risk patterns in new payment methods with associated amounts.

Ernst & Young (Data Science and Machine Learning Developer) Bangalore , India
(Consultant) Oct, 2025 - Feb, 2026

- SQL injection testing automation for DVWA using web crawling, machine learning and generative-AI payload generation; manual effort reduced 60%, scan time curbed 50%.
- Built and evaluated Responsible AI models to detect and classify toxicity, bias, and unsafe language in text data, increasing detection accuracy by 45% and improving overall NLP system safety and fairness.

(Associate Consultant) Aug, 2024 - Sep, 2025

- Developed LLM features in Python (PyTorch/TensorFlow) for code parsing, embedding-based graph generation, automated PDF/Excel output, and RAG SQL/vector DB integration, improving accuracy and automation by 60%.
- Built a quantitative risk-assessment system using probabilistic models, Monte Carlo simulation, and SQL-based threat intelligence to generate risk-adjusted mitigation strategies, improving analytical efficiency by 45
- Deployed an LLM-powered vulnerability scanner with transformer-based NLP and a conversational interface, improving real-time threat detection by 40%.

Hitachi (Research Intern) Bangalore, India
(Data Science and Machine Learning) Aug 2023 - Nov 2023

- Designed ML models (SVM, XGBoost, Linear Regression) for leakage detection by integrating sensor data (pressure, flow, temperature) with system topology, achieving 97.5% accuracy.
- Improved response time by 35% and lowered false positives by 60%, increasing predictive maintenance efficiency.

Skills

Programming & Tools:	Python, R, SQL, MATLAB, Git, PySpark, Flask, VS Code, Postman
ML & AI Expertise:	Machine Learning, Deep Learning, NLP, LLMs, Generative AI.
Databases & Visualization:	MongoDB, MySQL, Tableau, Power BI
Platforms & Design:	AWS, UI/UX Design, Figma.
Financial Skills:	Financial Modelling and Valuation , Financial Analysis
Professional Skills:	Critical & Analytical Thinking ,Problem Solving , Teamwork ,Adaptability

Projects

Finance AI Agent *Python, HTML, CSS, Flask* *Live Demo* | *GitHub*

- Engineered an AI assistant delivering real-time stock updates, financial insights, and market history, enhancing investment decisions with personalized recommendations. Implemented trend analysis and stock suggestion logic using Finhub API for a tailored user experience.

Order Book Liquidity Engine *Python, HTML, CSS, Flask* *Live Demo* | *GitHub*

- Developed an interactive limit-order-book analytics platform to analyze bid–ask spreads, market depth, order flow, and liquidity conditions using quantitative market-microstructure analysis.

Market Impact Simulator *Python, HTML, CSS, Flask* *Live Demo | GitHub*

- Built an interactive simulator to quantify trade execution costs, slippage, and market impact under varying trade sizes and liquidity conditions for algorithmic execution analysis..

Black-Scholes Pro *Python, HTML, CSS, Flask* *Live Demo | GitHub*

- Developed an interactive derivatives-pricing application implementing the Black-Scholes model to calculate option prices and analyze key option-pricing parameters and Greeks..

Volatility Terrain Survey *Python, HTML, CSS, Flask* *Live Demo | GitHub*

- Built an interactive volatility-analysis platform to visualize and analyze volatility patterns across different market conditions, maturities, and option-pricing scenarios.

Algorithmic Trading Battle Simulator *Python, HTML, CSS, Flask* *Live Demo | GitHub*

- Developed a trading simulation environment to evaluate and compare algorithmic trading strategies under dynamic market conditions using quantitative performance and risk metrics.

Systemic Risk Engine *Python, HTML, CSS, Flask* *Live Demo | GitHub*

- Built a quantitative risk-analysis engine to model interconnected financial exposures and assess systemic risk propagation across portfolios and financial entities.

Monte Carlo Wealth Simulator *Python, HTML, CSS, Flask* *Live Demo | GitHub*

- Developed a Monte Carlo simulation platform to model portfolio wealth trajectories under uncertain returns and evaluate long-term risk, growth, and probability-based outcomes.

Sentiment Analysis - Chrome Extension *Python, HTML, CSS, Javascript, Flask* *Live Demo | GitHub*

- Developed a sentiment analysis tool to identify sentiment keywords, compute text polarity, and visualize results through charts and word clouds; built an interactive front-end and packaged it as a Chrome extension for real-time text analysis.

PUBLICATION

- **Machine Learning based Spectrum Sensing and Spectrum Management Using Blockchain**
  International Journal of Scientific Research in Engineering and Management(IJSREM) Link | Certificate
- **Multi-Model Financial Fraud Detection With Explainable AI**
  International Journal For Multidisciplinary Research(IJFMR) Link | Certificate

EDUCATION

R.V. College Of Engineering Bangalore,India
B.E (Bachelor of Engineering) Electronics and Telecommunications *CGPA: 7.69*

Kendriya Vidyalaya No. 1 Raipur
Senior Secondary (Class XII) *Percentage: 85.6*

Kendriya Vidyalaya No. 1 Raipur
Secondary (Class X) *CGPA: 10*

CERTIFICATIONS

Google Data Analytics Coursera(Google)

EY Artificial Intelligence-Applied AI-Bronze Learning Badge Ernst & Young

Fundamentals of Financial Analysis Coursera(London Business School)

SEBI Investor Awarness Test SEBI and NISM

EXTRA-CURRICULAR

Teacher, Science (Voluntary) Agastya
Coordinator(Marketing and Public Relation) Entrepreneurship Cell
Cricket National and Regional Kendriya Vidyalaya
Swimming District (3rd) Raipur, Chhattisgarh